

Dev Patel

(+91) 8000722744 | devpatel0262@gmail.com 
linkedin.com/in/devpatel301  | github.com/devpatel301  | devpatel301.github.io 

EDUCATION

International Institute of Information Technology, Hyderabad
B.Tech in Computer Science

Hyderabad, India
July 2024 – Present

RELEVANT COURSEWORK

Computer Science: Data Structures & Algorithms, Operating Systems & Networks, Advanced Algorithms, Computer Systems, Automata Theory, Databases and Applications, Machine Learning, ML Security

Development: Design, Testing & Analysis of Software Systems, Internet of Things

Mathematics: Linear Algebra, Real Analysis, Probability and Statistics, Discrete Structures

EXPERIENCE

S-Learn (Learning Management System) | 
Shankar Foundation

January 2026 – April 2026
Hyderabad, India

- Led a 5-member team to develop a digital learning platform utilizing **Prototyping model** via **iterative client feedback**.
- Redesigned 70+ pages of static educational content into interactive, gamified web modules helping **100+ students**.
- Deployed production-ready releases on **Vercel** and **SPCRC servers (IIIT Hyderabad)** with **role-based access control** for teachers, parents, and admins.

PROJECTS

C-Shell (Unix Shell) | 

August 2025

- Developed a Unix-like shell with AST-based parsing, **pipelines**, and **job control**.
- Added 3 core controls: I/O redirection, background jobs, and signal handling for process management.

xv6 Demand Paging | 

September 2025 – October 2025

- Implemented **demand paging** in xv6 during the OSN mini-project (Sep 2025), enabling **lazy allocation** and on-demand loading for heap, stack, and executable accesses.
- Added **FIFO page replacement**, swap-in/swap-out support, and a **memstat** syscall, supporting up to **1024 swap pages (4MB per process)** and validating behavior across **6 xv6 test programs**.

Docs++ (Distributed Document Editor) | 

October 2025 – November 2025

- Built a collaborative document editor supporting **10+ concurrent users** with **sentence-level locking** over **TCP sockets**.
- Reduced metadata lookup latency to **sub-1ms** using **LRU caching** and **Trie-based indexing**.

Splitco (Expense Manager) | 

December 2025 – Present

- Built a 3-tier group expense splitting application using **React**, **Node.js**, and **MongoDB**.
- Reduced settlement complexity to at most n-1 edges using **graph-based debt netting** across n transactions.
- Designed a responsive UI with real-time updates across 4 modules: groups, expenses, balances, and settlements.

Agri-Bhasha Toolkit | Megathon 2025 | 

October 2025

- Collaborated with a 5-person team to engineer a multilingual agricultural assistant using **Bhashini API** with **OCR-based extraction** for **22 Indian languages**.
- Shipped a 3-service deployment stack across **Vercel** and **Render** managing **high-quality pull requests** and **peer code reviews**.

TECHNICAL SKILLS

Languages: C, C++, x86-64 Assembly, Python, HTML/CSS, JavaScript, TypeScript, SQL, Bash

Frameworks: React, Vite, Flask, Node.js, Express.js

Tools & Platforms: Git, GitHub, Wireshark, Docker, Postman, LaTeX, Figma, MongoDB, MySQL

Libraries: NumPy, Pandas, Matplotlib

ACHIEVEMENTS

- JEE Mains 2024 - AIR 1139 (Top 0.1%) | JEE Advanced 2024 - AIR 4199 (Top 2%).
- Top 20 Finalist, Megathon 2025 (Agri-Bhasha Toolkit) – IIIT H's largest hackathon with 1000+ participants.
- Marketing and Social Media Head, The Gaming Club – largest club at IIIT Hyderabad with 1200+ followers.
- Coordinated cultural activities for interhouse competitions as a captain of Prithvi House, Cultural Council.